

Course Syllabus Part II
DSC 670 – Applied Uses of Generative AI
3 Credit Hours

Course Resources

Course Text(s):

[*Generative AI in Action*](#)

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Required Resources:

In this course, you will need to be able to:

- Access to the Internet
- Access to Cyberactive
- Access to GitHub
- Microsoft Word
- Microsoft Excel
- Microsoft PowerPoint
- Jupyter Notebook

Course Schedule

Week	Topic	Assignment
1	Introduction to Generative AI (GAI)	Discussion/Participation Exercise
2	Generative AI Interfaces	Discussion/Participation Exercise Project Milestone 1: Pick Your Project
3	Working with Language Models	Discussion/Participation Exercise
4	Working with Image Generation Models	Discussion/Participation Exercise
5	Advanced Prompt Engineering	Discussion/Participation Exercise Project Milestone 2: Refine Your Project
6	Back to Reality – Retrieval Augmented Generation	Discussion/Participation Exercise
7	Tuning and Adapting LLMs Part I	Discussion/Participation Exercise

Week	Topic	Assignment
8	Tuning and Adapting LLMs Part II	Discussion/Participation Exercise
9	GAI Application Architectures	Discussion/Participation Exercise Project Milestone 3: Iterate Your Project
10	GAI Production Deployment	Discussion/Participation Exercise
11	Challenges and Security with Generative AI	Discussion/Participation Exercise
12	Ethical and Legal Considerations of Generative AI	Discussion/Participation Project Milestone 4: Create Your Application to Finish the Project!

Course Activities

In this section of the syllabus, I will describe what we will be doing in each of the activities for each week. Specifically, I will be describing your deliverables – those items you need to submit at or before the deadline. You can find more detail on grading criteria for each category by viewing its detailed rubric.

Written Assignments

Several weeks require a written assignment, or an assignment that starts using technology (e.g., prompt engineering) with a written explanation of the process followed and observations of the results. It's important that your narrative provide more than a simple explanation, specifically incorporating your critical observations.

Exercises

Every week you will be assigned one or more exercises based on the weekly topic to complete and submit to the assignment link. These are not group assignments to complete and should be done on your own. However, if you have questions about a specific assignment, you are encouraged to discuss via Teams or the discussion board.

Discussion

In accordance with the Discussion Board posting requirements outlined in Teams, there is a weekly discussion every week of the course. Each post (including replies to other students' posts) must be a minimum of 250 words and contain at least one credible source. These responses should be "substantive" which means more than, "Neat!" or "Good job!" They should also not contain jargon or be a post that boils down to you reposting the same thing you're commenting on in a different way.

Term Project

Over the course of the term, you will be working on a project proposal that has 4 milestones. Each milestone will be submitted separately with the final project submitted the last week of the course.

Week #	Written Assignment in Discussion Board	Discussion Responses	Exercise	Project Milestone
Week 1	X	X	X	
Week 2	X	X	X	X
Week 3	X	X	X	
Week 4	X	X	X	
Week 5	X	X	X	X
Week 6	X	X	X	
Week 7	X	X	X	
Week 8	X	X	X	
Week 9	X	X	X	X
Week 10	X	X	X	
Week 11	X	X	X	
Week 12	X	X		X

Grade and Point Breakdown

Component	Percentage	Point Value	Number of Submissions	Total Points
Discussion/Participation	34.6%	75 Points	12 submissions	900
Exercises	42.3%	100 Points	11 submissions	1100
Term Project Milestones	23.1%	150 Points	4 submissions	600
			Total Points	2600

Late Work

Late work is not accepted unless arrangements are made with the instructor for *very* special, unavoidable circumstances. If you do not alert the professor before or shortly after something that will make you late, the chances of special arrangements are much lower. If in doubt, please email as soon as possible.

Participation

Students are expected to login often and contribute to the class on a regular basis, including posting to the discussion board, submitting assignments, and participating in group activities as required. If you have specific participation requirements related to your educational funding or student status, you are expected to monitor your own participation to ensure you are in compliance with those requirements.

Expectations for Students

- Students should expect to spend approximately 10-15 hours per week to complete the activities and assignments in this course.

- Students will log in as often as needed to complete their assignments and progress through the course.
- Students will treat their classmates and the instructor with respect and courtesy.
- Students are responsible for keeping current with the reading assignments and coming to class prepared to discuss the work assigned.
- Students are responsible for knowing what assignments are due and when.
- Students will submit only their own work and will not commit plagiarism or other acts of academic dishonesty.
- Students will contact the instructor as soon as personal problems arise that may affect the student's ability to complete assignments on time.

Expectations for Faculty

- The instructor will treat all students with respect and courtesy.
- The instructor will make grading criteria clear and follow the criteria scrupulously in evaluating student work.
- The instructor will provide feedback about student work within 6 days of due dates (or 24 hours prior to the next due date)—feedback that helps the student learn and improve.
- The instructor will respond to all student messages within 48 hours.